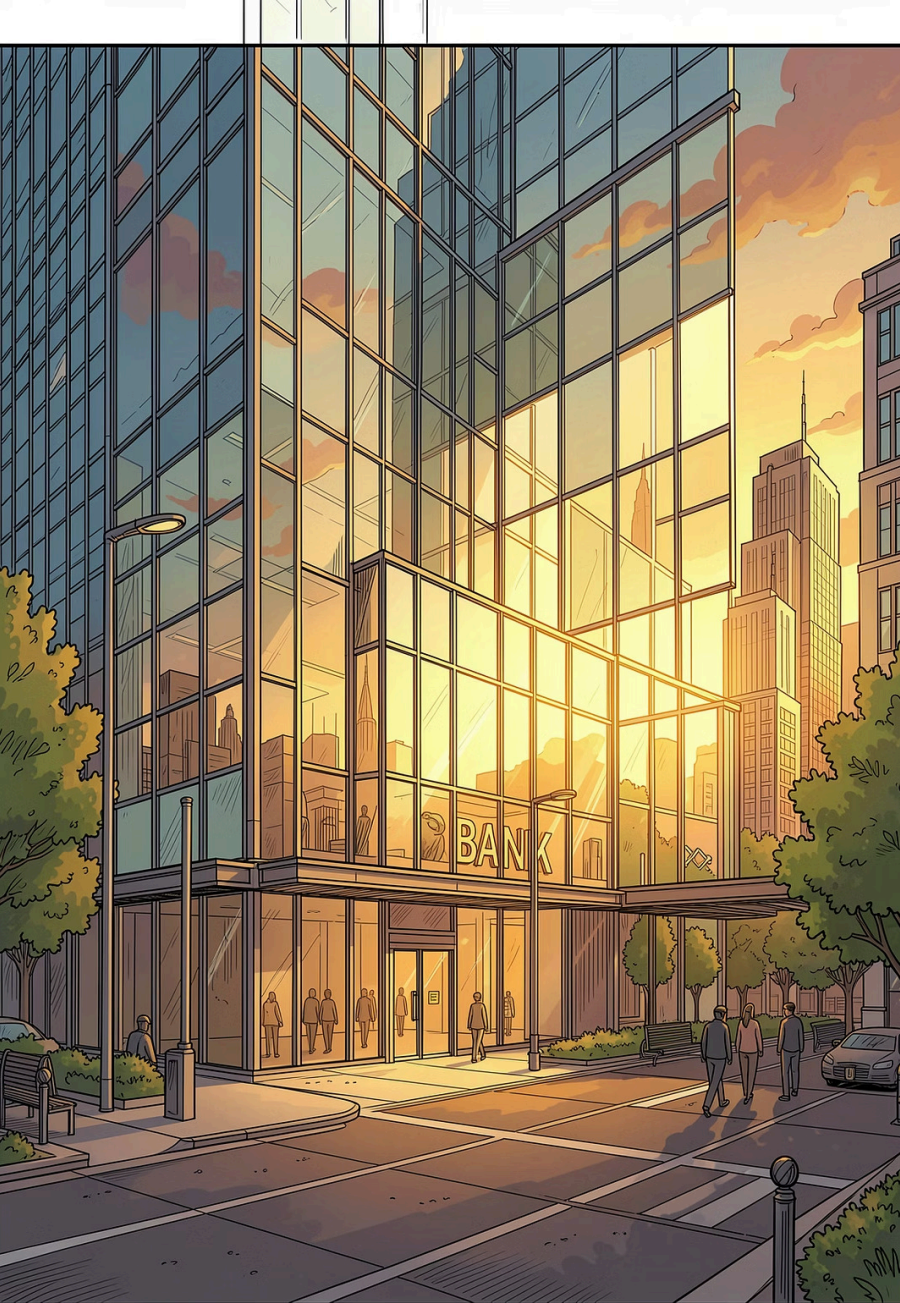




# Bank Transactions Analysis

Python · MySQL · Power BI

2,512 RECORDS · 16 COLUMNS

# Project Objectives

01

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## Clean & Preprocess

Handle missing values, standardize columns, engineer features

03

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## SQL Analysis

Extract business insights via MySQL queries

02

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## Explore (EDA)

Identify patterns in transactions, age, balance, channels

04

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## Visualize

Interactive Power BI dashboard for decision-making

# Tools & Workflow

## Python

Pandas, Seaborn, Matplotlib —  
cleaning & EDA

## MySQL

11 business queries for  
analytical insights

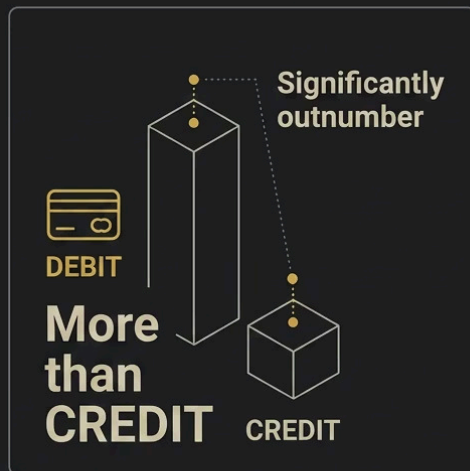
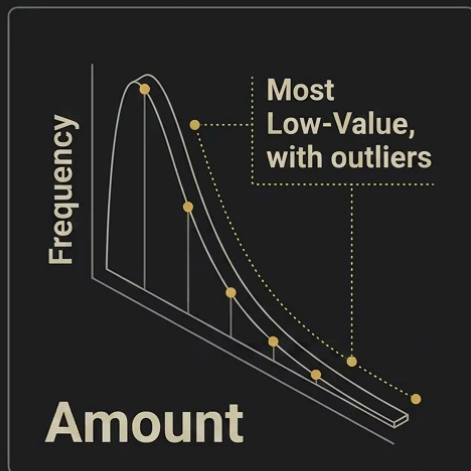
## Power BI

Interactive dashboard with KPIs & slicers





# Key Transaction Findings



## Debit Dominates

Customers primarily use banking for outgoing payments, not incoming credits

## Low-to-Medium Norm

Routine small transactions are the majority; high-value are rare

## Outliers Flagged

High-value transactions retained for premium/suspicious activity review

# Customer Demographics & Balance

## Adult & Middle-Aged

Highest financial activity and account balances

## Age → Balance

Older customers hold slightly higher balances ( $r = 0.32$ )

## Doctors Lead

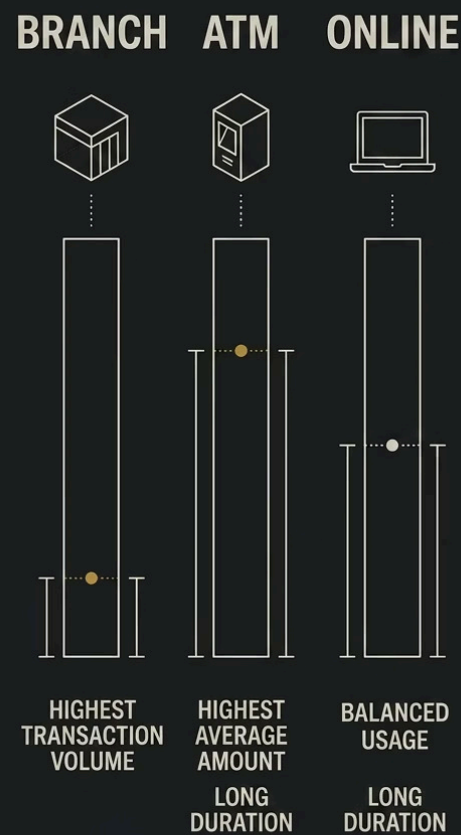
Doctor professionals maintain the highest average account balances

## Wealth ≠ Spend

High balance does not predict larger transaction amounts ( $r = -0.025$ )

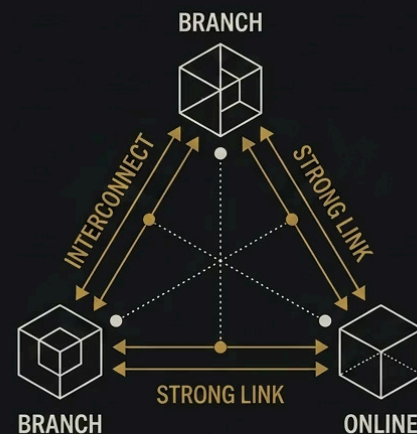
Young Adult and Senior groups also contributed higher **average** transaction values per SQL analysis.

# Channel Performance



## MULTI-CHANNEL ENGAGEMENT

### CHANNELS COMPARED



## STRONG INTERACTION

### CONSISTENT USAGE

## Multi-Channel Engagement

All three channels show significant usage — customers actively use both physical and digital banking.

⚠️ ATM & Online channels have the **longest transaction durations** — indicating lower operational efficiency.

Branch and ATM contributed the highest **total transaction value**. ATM holds the highest **average** transaction amount.

# Security & Login Insights



## Most Users: 1 Attempt

Single-attempt logins dominate — strong operational efficiency signal.

⊗ **Multiple login attempts** may indicate authentication issues or suspicious behavior — flagged for monitoring.

## High-Value + High Attempts

Transactions with unusually high login attempts require further security review.

# SQL Business Insights

1

## Top Accounts

AC00460 & AC00363 — highest transaction counts;  
candidates for premium retention

2

## Above-Average Spenders

Customers exceeding overall average transaction amount  
flagged for premium or suspicious review

3

## Balance Segmentation

Customers segmented by balance for targeted loan ads  
and marketing plans

4

## Rank by Age Group

Top-balance customers per age group identified for loan  
and marketing targeting



# Monthly Trends & Peak Activity



Transaction values fluctuated throughout the year, reflecting seasonal customer financial behavior.

## Mar-Apr

Significant drop in total transaction value

## Apr-May

Sharp recovery in activity

## September

Peak month — ~**\$73,000** total transactions



# Conclusions & Future Scope

## ✓ Delivered

End-to-end analytics: cleaning, EDA, SQL, Power BI dashboard

## 🔍 Key Insight

Debit dominates; doctors hold highest balances; September peaks at ~\$73K

## 🚀 Next Steps

Fraud detection ML · Customer segmentation · Transaction forecasting  
· Anomaly detection